

solution to secure national secrets, it's mainly intended to prevent criminals from getting access to, say, your credit card information off your stolen laptop. So having this feature in the age of portable devices like netbooks, MID's, etc, is certainly a huge boon.

IM conversation encryption

Most of us use instant messaging clients these days. We usually only chat with friends and relatives, so we don't realise the need to secure and encrypt our instant messages. But as the saying goes, "Better safe than sorry." So we should take steps to secure our IM chats—at least the somewhat-important conversations. And this is where OTR comes into the picture.

OTR (Off-the-Record) messaging helps us encrypt our conversations in a very simple manner. While it does provide most of the features offered by other similar protocols, like authentication and encryption, what makes it stand out is deniability. In fact, one of the primary goals behind OTR was to have deniability, much like in real-life conversations. So if you use OTR, the messages sent by you have no digital signature. Hence, someone can forge the identity of anyone else, once the conversation is over. You can deny your participation in the conversation. But when the conversation is on, we can be sure of the other person's identity. Another reason why OTR shines is that it is not tied to any single protocol. It's just that the users on both the ends must have clients that support OTR.

Many IM clients support OTR, either natively or by using a plug-in. On Linux, both our favourite IM clients, Pidgin and Kopete, support OTR. To use OTR in Pidgin, enable the 'Off-the-Record Messaging' plug-in under *Tools→Plugins*. Then, in its configuration menu, click 'Generate for each account' along with which OTR is to be used. Then open a chat window using the account on which OTR is enabled. A new option will be available in the window as shown in Figure 2. Here, we can choose whether we want to set the conversation to 'Private' or not. And if the person on the other side also has OTR support, then we are all set to go.

Incidentally, developers of Empathy, which is now the default IM client in the latest versions of Ubuntu, Fedora and most other distributions, are not too impressed with OTR. They plan on implementing XMPP end-to-end encryption. Though an OTR implementation is also being worked upon, primarily to have compatibility with other OTR users.

Internet browsing security

While these days, Internet surfing is probably the primary activity on our computing devices, we hardly ever take steps to protect our privacy and security on the Web. One of the first things that comes to mind when thinking about surfing with privacy, is the use of proxy servers.

Some of the proxy servers do help in surfing the Web securely and, to some extent, anonymously. But the main problem with using these kinds of proxy servers is that they

MICROPROCESSOR BASED BATTERY-CHARGER HEALTH MONITORS-AC/DC MODEL	MICROPROCESSOR BASED CHARGER SENSOR UNIT	BATTERY LIFE ENHANCER
 TO SENSOR UNIT TO SENSOR UNIT Computer interface AC POWER	 Sensor Units for Battery & Chargers	
• Microprocessor Based • Senses failures of charges in conjunction with sensor units. • Displays working of transformer along with output voltage • Displays working of Battery with voltage • Audio-visual alarm in case of failures of A/C Voltage failure as well as low/high battery voltage. • Different sensors of different voltage can be connected to same console • Computer port for attaching computer for data downloading • Input Voltage AC 110/220 or DC 12V Power supply based • Current draw 75mA @ 10%	• Microprocessor Based • Would be able to sense A/C failures in respect to A/C supply cable, step down transformer failure as well as fuse failure in A/C Circuits • Units available for 4V-6V-12-24-26-28-30-32-48-60—110 AC-DC Circuits • Dual LED Visual display for status • Unique alphanumeric 4 Digit I/D for each unit • I/D is programmable with ID Editor	Anti-sulphation device Increases Lead Acid Battery Life • Stops sulphation • Eliminates already existing sulphation • Increases battery lifetime by 200% • Reduces Electricity Bill as it keeps battery plates clean, which reduces recharge time • Helps protect the environment
PEARL TELECOMMUNICATION & ELECTRONICS (P) LTD.		
252, Kailash Plaza (LGF), Sant Nagar, East of Kailash, New Delhi-110065 Call: 9650746663, 9810001552 E-mail: info@pearltele.com, Website: www.pearltele.com		